

PHY101

LEC 1 TO 22

FORMULAS

BY VU BSCS

Lecture 1: Introduction to Physics and This Course

- Basic concepts:
 - Units and measurements: SI units (meter, kilogram, second, ampere, kelvin, mole, candela).
 - Vector and scalar quantities.

Lecture 2: Kinematics – I

- Displacement, Velocity, and Acceleration:
 - Displacement: $\vec{d} = \vec{r}_f - \vec{r}_i$
 - Velocity: $\vec{v} = \frac{d\vec{r}}{dt}$
 - Acceleration: $\vec{a} = \frac{d\vec{v}}{dt}$
- Equations of Motion (constant acceleration):
 - $v = u + at$
 - $s = ut + \frac{1}{2}at^2$
 - $v^2 = u^2 + 2as$

Lecture 3: Kinematics – II

- Projectile motion:
 - Horizontal motion: $x = v_0 t$
 - Vertical motion: $y = v_0 t - \frac{1}{2} g t^2$
 - Range: $R = \frac{v_0^2 \sin(2\theta)}{g}$
- Circular motion:
 - Angular velocity: $\omega = \frac{\Delta\theta}{\Delta t}$
 - Centripetal acceleration: $a_c = \frac{v^2}{r}$

Lecture 4: Force and Newton's Laws

- Newton's Laws of Motion:
 - First law (Inertia): $\vec{F} = 0 \Rightarrow$ constant velocity
 - Second law (Force and Acceleration): $\vec{F} = m\vec{a}$
 - Third law (Action and Reaction): $\vec{F}_{12} = -\vec{F}_{21}$
- Gravitational force:
 - $F_g = mg$

Lecture 5: Applications of Newton's Laws – I

- Friction:
 - Static friction: $f_s \leq \mu_s N$
 - Kinetic friction: $f_k = \mu_k N$
- Inclined plane:
 - Parallel force: $F_{\parallel} = mg \sin \theta$
 - Perpendicular force: $F_{\perp} = mg \cos \theta$

Lecture 6: Applications of Newton's Laws – II

- Tension in a rope (e.g., for an object in equilibrium):
 - $T = mg$ (if the object is hanging vertically)
- Atwood's Machine (two masses):
 - $T = \frac{2m_1m_2g}{m_1 + m_2}$

Lecture 7: Work and Energy

- Work:
 - $W = \vec{F} \cdot \vec{d} = Fd \cos \theta$
- Kinetic Energy:
 - $KE = \frac{1}{2}mv^2$
- Potential Energy:
 - $PE = mgh$
- Work-Energy Theorem:
 - $W = \Delta KE$

Lecture 8: Conservation of Energy

- Conservation of mechanical energy:
 - $E_{\text{total}} = KE + PE$ (constant if no non-conservative forces are involved)
- Power:
 - $P = \frac{W}{t} = Fv$

Lecture 9: Momentum

- Momentum:
 - $\vec{p} = m\vec{v}$
- Impulse:
 - $\vec{J} = \Delta\vec{p} = F\Delta t$
- Conservation of Momentum:
 - $\vec{p}_{\text{initial}} = \vec{p}_{\text{final}}$

Lecture 10: Collisions

- Elastic Collision:
 - $m_1v_1 + m_2v_2 = m_1v'_1 + m_2v'_2$
- Inelastic Collision:
 - Kinetic energy is not conserved, but momentum is.

Lecture 11: Rotational Kinematics

- Angular displacement, velocity, and acceleration:
 - $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$
 - $\omega = \omega_0 + \alpha t$
- Rotational Kinetic Energy:
 - $KE_{\text{rot}} = \frac{1}{2}I\omega^2$

Lecture 12: Physics of Many Particles

- Center of Mass:
 - $\vec{R}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$
 - System of particles:
 - Conservation of momentum: $\sum \vec{p} = \text{constant}$
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Lecture 13: Angular Momentum

- Angular momentum:
 - $\vec{L} = \vec{r} \times \vec{p}$
- Conservation of angular momentum:
 - $L_{\text{initial}} = L_{\text{final}}$

Lecture 14: Equilibrium of Rigid Bodies

- Conditions for equilibrium:
 - Translational equilibrium: $\sum \vec{F} = 0$
 - Rotational equilibrium: $\sum \vec{\tau} = 0$
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Lecture 15: Oscillations – I

- Simple Harmonic Motion (SHM):
 - $x(t) = A \cos(\omega t + \phi)$
 - $a = -\omega^2 x$
- Period of SHM:
 - $T = 2\pi\sqrt{\frac{m}{k}}$ (for mass-spring system)

Lecture 16: Oscillations – II

- Damped Oscillations:
 - $x(t) = Ae^{-\gamma t} \cos(\omega' t + \phi)$
 - Forced Oscillations:
 - $x(t) = \frac{F_0}{m} \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}$
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Lecture 17: Physics of Materials

- Stress and Strain:
 - Stress: $\sigma = \frac{F}{A}$
 - Strain: $\varepsilon = \frac{\Delta L}{L}$
- Young's Modulus:
 - $Y = \frac{\sigma}{\varepsilon}$

Lecture 18: Physics of Fluids

- Continuity Equation:
 - $A_1 v_1 = A_2 v_2$
- Bernoulli's Equation:
 - $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$
- Hydrostatic Pressure:
 - $P = \rho gh$

Lecture 19: Physics of Sound

- Wave speed:
 - $v = \sqrt{\frac{B}{\rho}}$ (for longitudinal waves in a medium)
 - Doppler Effect:
 - $f' = f \left(\frac{v \pm v_o}{v \pm v_s} \right)$
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Lecture 20: Wave Motion

- Wave equation:
 - $y(x, t) = A \sin(kx - \omega t + \phi)$
- Wave speed:
 - $v = \lambda f$

Lecture 21: Gravitation

- Gravitational Force:
 - $F = G \frac{m_1 m_2}{r^2}$
- Gravitational Potential Energy:
 - $U = -\frac{GMm}{r}$
- Kepler's Third Law:
 - $T^2 = \frac{4\pi^2}{GM} r^3$

Lecture 22: Electrostatics – I

- Coulomb's Law:

- $F = k_e \frac{q_1 q_2}{r^2}$

- Electric Field:

- $\vec{E} = \frac{\vec{F}}{q}$

- Gauss's Law:

- $\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$

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